

THE CONSTITUTION OF POCKETDOCTOR MARK 4

The Sovereign Foundational Charter, Articles, Schedules, and Technical Ordinances Governing the ESP32 AI-Powered Handheld Diagnostic Assistant

PREAMBLE

WE, THE ARCHITECTS AND DEVELOPERS OF EDGE HEALTH INTELLIGENCE, having solemnly resolved to constitute PocketDoctor Mark 4 into a Sovereign, Autonomous, AI-Powered Point-of-Care Health Triage System, securing for all users Instant Health Assessment, Absolute Data Persistence, Privacy of Local Memory Storage, and Unrestricted Access to Cloud Intelligence, DO HEREBY ADOPT, ENACT, AND GIVE TO THIS SYSTEM THIS CONSTITUTION.

PART I: THE SYSTEM AND ITS HARDWARE BOUNDARIES

Article 1: Name and Primary Function

- This firmware system shall be formally known as **PocketDoctor Mark 4** (running core v3.0 architecture).
- Its primary function is established as an autonomous, handheld, point-of-care medical triage assistant capable of physical menu interaction, local profile persistence, embedded web server management, and multi-turn LLM clinical reasoning.

Article 2: Hardware Supremacy

- The hardware foundation shall be powered by an **Espressif ESP32** dual-core 32-bit Xtensa LX6 microcontroller operating at 240 MHz.
- Visual rendering is mandated to an **SSD1306 OLED Display** (128x64 pixels, monochrome) communicating over I2C at address 0x3C.
- User navigation and physical input shall be governed by a **KY-023 Dual-Axis Analog Joystick** equipped with an integrated push-button switch.

Article 3: Immutable Signal Topology

- The GPIO pin allocations defined in the **First Schedule** are hereby declared immutable hardware lines.
- I2C Data (SDA) on **GPIO 21** and I2C Clock (SCL) on **GPIO 22** shall serve the SSD1306 OLED display.
- Analog inputs VRx (**GPIO 34**) and VRy (**GPIO 35**) alongside digital push-button SW (**GPIO 32**, INPUT_PULLUP) shall govern joystick input.

PART II: PERSISTENT MEMORY AND PATIENT REGISTRY

Article 4: The EEPROM Memory Realm

- A non-volatile EEPROM memory block of exactly **512 Bytes** (#define EEPROM_SIZE 512) is set aside for system configurations and patient registry storage.

Article 5: Magic Key Authentication

- Byte address 0x000 shall contain the 2-byte Magic Key Identifier (0xABCD).
- No stored Wi-Fi configuration shall be deemed valid unless verified against this Magic Key. If invalid, the system shall fall back to default firmware credentials (NKJIO2.4).

Article 6: Patient Profiles Directive

- Up to **three (3) individual patient profiles** may be persisted inside EEPROM starting at byte address 100.
- Each profile record shall encapsulate seven immutable attributes: Name, Age (1–120), Height (50–250 cm), Weight (20–300 kg), Intoxication Status (None, Alcohol, Tobacco, Both), Medical History Flag, and Preferred Treatment Modality (Allopathy, Homeopathy,

Ayurvedic, or All types).

PART III: SYSTEM EXECUTION AND DYNAMIC STATE MACHINE

Article 7: The Non-Blocking Principle

1. The core loop (loop()) shall execute on a non-blocking asynchronous principle.
2. Processing background web server requests (server.handleClient()) shall take place continuously regardless of physical UI state.

Article 8: The Nine System States

The system lifecycle shall transition strictly between nine execution states:

- **S₀ (Boot State):** Peripherals initialization, Serial setup at 115200 baud, EEPROM read, and Wi-Fi connection.
- **S₁ (Web Active State):** Port 80 Web Server listening for network client requests.
- **S₂ (Profile Menu State):** Interactive list of saved patient profiles or option to create new.
- **S₃ (Profile Builder State):** On-device multi-step form builder with letter-scrolling and numeric adjusters.
- **S₄ (Symptom Selection State):** Multi-select checklist of 35 standard medical symptoms.
- **S₅ (Duration Selection State):** Duration picker (1-3 days, 4-7 days, 1-2 weeks, >2 weeks).
- **S₆ (Groq AI Q&A; State):** 10-turn dynamic Yes/No adaptive dialogue via Groq Cloud API.
- **S₇ (OLED Diagnosis Render State):** Short diagnosis summary card rendered on OLED.
- **S₈ (HTML Report State):** Comprehensive CSS-styled medical report generated and hosted at /report.

PART IV: THE SUPREME AI EXECUTIVE AND GROQ API

Article 9: The Groq Inference Mandate

1. Clinical reasoning authority is delegated to Groq Cloud's llama-3.3-70b-versatile model over HTTPS.

Article 10: The Ten Questions Doctrine

1. During diagnostic evaluation, the AI shall issue exactly **ten (10) consecutive Yes/No questions** (maximum 10 words each) tailored to the patient's symptoms and profile context.

Article 11: Structured Output Extraction

1. Upon completing 10 turns, the AI shall provide output matching the exact format: Diagnosed Disease, Accuracy %, Medicines, Precautions, and Critical Flag.
2. Extracted metrics populate both the OLED screen and the full HTML medical report.

PART V: THE GOVERNANCE PORTAL AND WEB SERVER

Article 12: Port 80 Governance

1. An embedded HTTP Web Server shall run on Port 80.
2. All web pages shall be stored in flash memory using PROGMEM blocks to conserve SRAM.

Article 13: Web Portal User Rights

1. Connecting users shall enjoy the right to view system status (/ & /portal), manage Wi-Fi settings (/wifi), execute profile web CRUD operations (/profiles), and view/download complete medical reports (/report).

FIRST SCHEDULE: MASTER HARDWARE PINOUT MATRIX

Component	Signal Pin	ESP32 GPIO	Specifications & Protocols
SSD1306 OLED	SDA	GPIO 21	I2C Serial Data (3.3V Logic)
SSD1306 OLED	SCL	GPIO 22	I2C Serial Clock (Address 0x3C)
KY-023 Joystick	VRx	GPIO 34	Analog X-Axis (12-bit ADC1_CH6)
KY-023 Joystick	VRy	GPIO 35	Analog Y-Axis (12-bit ADC1_CH7)
KY-023 Joystick	SW	GPIO 32	Digital Push Button (INPUT_PULLUP)

First Schedule Table: Constitutional Hardware Assignment Matrix

SECOND SCHEDULE: EEPROM MEMORY MAP

Address Range	Segment Size	Data Segment Description & Encoding
0x000 - 0x001	2 Bytes	Magic Key Verification Identifier (0xABCD)
0x002 - 0x022	33 Bytes	Stored Wi-Fi SSID (Null-Terminated String)
0x023 - 0x043	33 Bytes	Stored Wi-Fi Password (Null-Terminated String)
0x044 - 0x063	32 Bytes	Reserved System Memory Space
0x064 - 0x1FF	412 Bytes	Profile Storage Segment (Count + Up to 3 Serialized Profiles)

Second Schedule Table: Non-Volatile EEPROM Memory Map

THIRD SCHEDULE: WEB SERVER ROUTE MATRIX

Endpoint Route	Method	Handler Function	Administrative Description
/ & /portal	GET	handleDashboard()	Main system dashboard with local IP & uptime
/wifi	GET	handleWifi()	Wi-Fi network configuration page
/wifi/update	POST	handleWifiUpdate()	Saves new credentials to EEPROM & reboots
/profiles	GET	handleProfiles()	Patient profiles web management list
/profile/edit	GET	handleProfileEdit()	Web form for editing or adding profiles
/profile/update	POST	handleProfileUpdate()	Validates parameters and commits to EEPROM
/profile/delete	GET	handleProfileDelete()	Deletes profile at given index
/report	GET	handleReport()	Hosts responsive HTML clinical report

Third Schedule Table: HTTP Web Server RESTful Route Matrix

FOURTH SCHEDULE: MEDICAL DISCLAIMER & ORDINANCES

1. **Non-Clinical Disclaimer:** PocketDoctor Mark 4 is an experimental AI research prototype. Outputs do not constitute clinical diagnosis or professional medical advice.
2. **Attestation & Licensing:** This Constitution and system specification is licensed under the MIT License and attributed to **Vaidik Khurana**.

ENACTED, RATIFIED, AND COMMITTED TO NON-VOLATILE FLASH

By Order of the Lead Architect: **Vaidik Khurana**

PocketDoctor Mark 4 Development Group | Date: July 2026